

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250279855

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Okano; Mayuko et al.

TERMINAL AND COMMUNICATION METHOD

Abstract

A terminal includes a receiving unit configured to receive, from a base station, a downlink shared channel in a virtual CC (component carrier) that is discontinuous in a frequency domain and that includes a plurality of actual CCs, a control unit configured to determine feedback information related to retransmission control corresponding to the downlink shared channel, and a transmitting unit configured to transmit, to the base station, an uplink control channel including the feedback information in the virtual CC, wherein the control unit determines which actual CC included in the virtual CC transmits the uplink control channel, and determines an interval between the downlink shared channel and the uplink control channel.

Inventors: Okano; Mayuko (Chiyoda-ku, Tokyo, JP), Kumagai; Shinya (Chiyoda-ku, Tokyo, JP), Yoshioka; Shohei (Chiyoda-ku, Tokyo, JP), Takahashi; Yuki (Chiyoda-ku, Tokyo, JP), Okamura; Masaya (Chiyoda-ku, Tokyo, JP)

Applicant: NTT DOCOMO, INC. (Tokyo, JP)

Family ID: 88518130

Assignee: NTT DOCOMO, INC. (Tokyo, JP)

Appl. No.: 18/857636

Filed (or PCT Filed): April 25, 2022

PCT No.: PCT/JP2022/018765

Publication Classification

Int. Cl.: H04L1/1812 (20230101); H04L5/00 (20060101)

U.S. Cl.:

Background/Summary

TECHNICAL FIELD

[0001] The present invention relates to terminals and communication methods in a wireless communication system.

BACKGROUND ART

[0002] In new radio (NR) (also referred to as “5G”) which is an advanced system succeeding long term evolution (LTE), a study has been conducted on a technology that satisfies required conditions such as a large-capacity system, high-speed data transmission speed, short delay, simultaneous connection to a large number of terminals, low cost, power saving, and the like (for example, Non-Patent Literature 1).

[0003] Furthermore, a study on 6G has been started as a post-5G next-generation wireless communication system, and it is expected to realize wireless quality exceeding 5G. For example, in 6G, a study has been conducted to realize further increase in capacity, use of a new frequency band, further reduction in delay, higher reliability, further reduction in power consumption, extension of coverage in new areas (high altitudes, sea, and outer space) by a non-terrestrial network, and the like (for example, Non-Patent Literature 2).

CITATION LIST

Non-Patent Literature

[0004] Non-Patent Literature 1: 3GPP TS 38.300 V16.8.0 (2021 December) [0005] Non-Patent Literature 2: 3GPP TR 38.821 V16.1.0 (2021 May) [0006] Non-Patent Literature 3: 3GPP TS 38.331 V16.7.0 (2021 December)

SUMMARY OF INVENTION

Technical Problem

[0007] A function to use a wideband to secure a data resource, such as carrier aggregation, is supported. When using the carrier aggregation, data resources need to be scheduled for respective carriers, so that a study on more flexible and efficient resource allocation is underway. However, a feedback operation related to retransmission control has not been stipulated in a wideband that is discontinuous in the frequency domain.

[0008] The present invention has been made in view of the above points, and an object of the present invention is to execute a feedback operation related to retransmission control in a wideband that is discontinuous in the frequency domain in a wireless communication system.

Solution to Problem

[0009] According to the disclosed technology, there is provided a terminal including a receiving unit configured to receive, from a base station, a downlink shared channel in a virtual CC (component carrier) that is discontinuous in a frequency domain and that includes a plurality of actual CCs, a control unit configured to determine feedback information related to retransmission control corresponding to the downlink shared channel, and a transmitting unit configured to transmit, to the base station, an uplink control channel including the feedback information in the virtual CC, wherein the control unit determines which actual CC included in the virtual CC transmits the uplink control channel, and determines an interval between the downlink shared channel and the uplink control channel.

Advantageous Effects of Invention

[0010] According to the disclosed technology, in a wireless communication system, a feedback

operation related to retransmission control can be executed in a wideband that is discontinuous in the frequency domain.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0011] FIG. **1** is a diagram illustrating an example configuration (1) of a wireless communication system.

[0012] FIG. **2** is a diagram illustrating an example configuration (2) of the wireless communication system.

[0013] FIG. **3** is a diagram illustrating an example (1) of a configuration of a virtual CC according to an embodiment of the present invention.

[0014] FIG. **4** is a diagram illustrating an example (2) of the configuration of the virtual CC according to the embodiment of the present invention.

[0015] FIG. **5** is a diagram illustrating an example of a type-1 HARQ-ACK codebook.

[0016] FIG. **6** is a diagram illustrating an example of a type-2 HARQ-ACK codebook.

[0017] FIG. **7** is a diagram illustrating an example of a PUCCH group.

[0018] FIG. **8** is a diagram illustrating an example (1) of a PUCCH cell according to the embodiment of the present invention.

[0019] FIG. **9** is a diagram illustrating an example (2) of the PUCCH cell according to the embodiment of the present invention.

[0020] FIG. **10** is a diagram illustrating an example (3) of the PUCCH cell according to the embodiment of the present invention.

[0021] FIG. **11** is a diagram illustrating an example (4) of the PUCCH cell according to the embodiment of the present invention.

[0022] FIG. **12** is a diagram illustrating an example (1) of an HARQ-ACK feedback timing according to the embodiment of the present invention.

[0023] FIG. **13** is a diagram illustrating an example (2) of the HARQ-ACK feedback timing according to the embodiment of the present invention.

[0024] FIG. **14** is a diagram illustrating an example (1) of the type-1 HARQ-ACK codebook according to the embodiment of the present invention.

[0025] FIG. **15** is a diagram illustrating an example (2) of the type-1 HARQ-ACK codebook according to the embodiment of the present invention.

[0026] FIG. **16** is a diagram illustrating an example (3) of the type-1 HARQ-ACK codebook according to the embodiment of the present invention.

[0027] FIG. **17** is a diagram illustrating an example (4) of the type-1 HARQ-ACK codebook according to the embodiment of the present invention.

[0028] FIG. **18** is a diagram illustrating an example (5) of the type-1 HARQ-ACK codebook according to the embodiment of the present invention.

[0029] FIG. **19** is a diagram illustrating an example (1) of the type-2 HARQ-ACK codebook according to the embodiment of the present invention.

[0030] FIG. **20** is a diagram illustrating an example (2) of the type-2 HARQ-ACK codebook according to the embodiment of the present invention.

[0031] FIG. **21** is a diagram illustrating an example (3) of the type-2 HARQ-ACK codebook according to the embodiment of the present invention.

[0032] FIG. **22** is a diagram illustrating an example (4) of the type-2 HARQ-ACK codebook according to the embodiment of the present invention.

[0033] FIG. **23** is a diagram illustrating an example of the functional configuration of a base station **10** according to the embodiment of the present invention.

[0034] FIG. **24** is a diagram illustrating an example of the functional configuration of a terminal **20** according to the embodiment of the present invention.

[0035] FIG. **25** is a diagram illustrating an example of the hardware configuration of the base station **10** or the terminal **20** according to the embodiment of the present invention.

[0036] FIG. **26** is a diagram illustrating an example configuration of a vehicle **2001** according to the embodiment of the present invention.

DESCRIPTION OF EMBODIMENTS

[0037] Hereinafter, an embodiment of the present invention will be described with reference to the drawings. It is noted that the embodiment described below is one example, and an embodiment to which the present invention is applied is not limited to the following embodiment.

[0038] In the operation of a wireless communication system of the embodiment of the present invention, an existing technology is used as appropriate. It may be noted that the existing technology is, for example, existing LTE, but is not limited to the existing LTE. In addition, the term “LTE” used in the present specification has a broad meaning including LTE-Advanced and post-LTE-Advanced systems (for example, NR), unless otherwise specified.

[0039] Further, in the embodiment of the present invention described below, terms such as a synchronization signal (SS), a primary SS (PSS), a secondary SS (SSS), a physical broadcast channel (PBCH), a physical random access channel (PRACH), a physical downlink control channel (PDCCH), a physical downlink shared channel (PDSCH), a physical uplink control channel (PUCCH), and a physical uplink shared channel (PUSCH) used in the existing LTE are employed. This is for the convenience of description, and substantially equivalent signals, functions, and the like may as well be referred to by other names. In addition, in the case of NR, the above-described terms correspond to NR-SS, NR-PSS, NR-SSS, NR-PBCH, NR-PRACH, and the like. It may be noted, however, that a signal is not necessarily specified as “NR-” even when it is specifically used in NR.

[0040] In the embodiment of the present invention, a duplex system may be a time division duplex (TDD) system, a frequency division duplex (FDD) system, or other (for example, flexible duplex or the like) systems.

[0041] In addition, in the embodiment of the present invention, a description indicating that a radio parameter or the like is “configured” may mean that a predetermined value is pre-configured, or may mean that a radio parameter notified from a base station **10** or a terminal **20** is configured.

[0042] FIG. **1** is a diagram illustrating an example configuration (1) of a wireless communication system according to the embodiment of the present invention. As illustrated in FIG. **1**, the wireless communication system according to the embodiment of the present invention includes a base station **10** and a terminal **20**. Although one base station and one terminal are illustrated in FIG. **1**, this is one example, and a plurality of base stations **10** and a plurality of terminals **20** may be provided.

[0043] The base station **10** is a communication device that provides one or more cells and performs wireless communication with the terminal **20**. Physical resources of a radio signal are defined in a time domain and a frequency domain, the time domain may be defined by the number of orthogonal frequency division multiplexing (OFDM) symbols, and the frequency domain may be defined by the number of sub-carriers or the number of resource blocks. The base station **10** transmits a synchronization signal and system information to the terminal **20**. The synchronization signal is, for example, NR-PSS and NR-SSS. The system information is transmitted through, for example, NR-PBCH and is also referred to as notification information. The synchronization signal and the system information may be referred to as an SSB (SS/PBCH block). As illustrated in FIG. **1**, the base station **10** transmits a control signal or data to the terminal **20** through a downlink (DL) and receives a control signal or data from the terminal **20** through an uplink (UL). Both the base station **10** and the terminal **20** can transmit and receive a signal by performing beamforming. In addition, both the base station **10** and the terminal **20** can apply multiple-input multiple-output

(MIMO) communication to the DL or the UL. In addition, both the base station **10** and the terminal **20** may perform communication via a secondary cell (SCell) and a primary cell (PCell) by carrier aggregation (CA). Furthermore, the terminal **20** may perform communication via a primary cell of the base station **10** and a primary secondary cell group cell (Primary SCG Cell, PSCell) of another base station **10** by dual connectivity (DC).

[0044] The terminal **20** is a communication device having a wireless communication function such as a smartphone, a mobile phone, a tablet, a wearable terminal, or a machine-to-machine (M2M) communication module. As illustrated in FIG. **1**, the terminal **20** uses various communication services provided by the wireless communication system by receiving a control signal or data from the base station **10** using the DL and transmitting a control signal or data to the base station **10** using the UL. In addition, the terminal **20** receives various reference signals transmitted from the base station **10** and measures propagation path quality based on the result of reception of the reference signals.

[0045] The terminal **20** can perform carrier aggregation for bundling a plurality of cells (a plurality of component carriers (CCs)) and performing communication with the base station **10**. In the carrier aggregation, one primary cell (PCell) and one or more secondary cells (SCells) are used. In addition, a PUCCH-SCell having a PUCCH may be used.

[0046] FIG. **2** is a diagram illustrating an example (2) of the wireless communication system according to the embodiment of the present invention. FIG. **2** illustrates an example configuration of the wireless communication system in a case where dual connectivity (DC) is executed. As illustrated in FIG. **2**, a base station **10A** serving as a master node (MN) and a base station **10B** serving as a secondary node (SN) are provided. Each of the base station **10A** and the base station **10B** is connected to a core network. The terminal **20** can communicate with both the base station **10A** and the base station **10B**.

[0047] A cell group provided by the base station **10A**, which is the MN, is referred to as a master cell group (MCG), and a cell group provided by the base station **10B**, which is the SN, is referred to as a secondary cell group (SCG). In addition, in the DC, the MCG includes one PCell and one or more SCells, and the SCG includes one primary SCG Cell (PSCell) and one or more SCells.

[0048] Operations for processing in the present embodiment may be executed in the system configuration illustrated in FIG. **1**, may be executed in the system configuration illustrated in FIG. **2**, or may be executed in a system configuration other than the above-described system configurations.

[0049] In addition, in LTE or NR, a carrier aggregation function using a wideband to secure data resources is supported. In the carrier aggregation function, a plurality of component carriers is aggregated, thereby making it possible to secure wideband data resources. For example, a 100 MHz bandwidth may be used by bundling a plurality of 20 MHz bandwidths.

[0050] In a conventional carrier aggregation function, there is a problem in that a data resource needs to be scheduled for each of a plurality of bundled component carriers, and resource allocation overheads are large.

[0051] Therefore, a description will be given as to a method of performing resource allocation in the unit of scheduling having a granularity different from that of the component carrier and a terminal that performs resource allocation in the unit of scheduling having a granularity different from that of the component carrier.

[0052] A framework that performs scheduling or aggregation at a granularity different from that of the component carrier is defined as frequency fragmentation.

[0053] In addition, in carrier aggregation, aggregation of a granularity different from that of the component carrier is defined as non-continuous carrier aggregation.

[0054] In addition, in the carrier aggregation (non-continuous carrier aggregation), performing scheduling at a granularity different from that of the component carrier is defined as non-continuous scheduling.

[0055] The granularity different from that of the component carrier described above may be the unit of virtual CCs, the unit of bandwidth parts (BWPs), the unit of physical resource blocks (PRBs), or the unit of PRB sets.

[0056] Here, a virtual CC is a carrier set in which all or some frequency resources included in each component carrier are bundled across the plurality of component carriers. For example, it may be assumed that the virtual CC includes a plurality of BWPs.

[0057] FIG. 3 is a diagram illustrating an example (1) of a configuration of the virtual CC according to the embodiment of the present invention. A virtual CC #i illustrated in FIG. 3 is a carrier set in which a BWP #a and a BWP #b included in respective component carriers are bundled across the plurality of component carriers (CC #0 and CC #1). Hereinafter, the CC #0 and the CC #1 are referred to as actual CCs.

[0058] Furthermore, it may be assumed that the virtual CC includes a plurality of PRBs or PRB sets.

[0059] FIG. 4 is a diagram illustrating an example (2) of the configuration of the virtual CC according to the embodiment of the present invention. A virtual CC #i illustrated in FIG. 4 is a carrier set obtained by bundling a plurality of PRBs included in respective component carriers across the plurality of component carriers (CC #0 and CC #1). It is noted that the plurality of PRBs or the PRB sets may be included in one or a plurality of BWPs.

[0060] The terminal 20 may transmit terminal capability information indicating the configuration of the virtual CC to the base station 10. The terminal capability information indicating the configuration of the virtual CC may be, for example, information indicating that the virtual CC includes a plurality of BWPs or information indicating that the virtual CC includes a plurality of PRBs.

[0061] Furthermore, the terminal capability information indicating the configuration of the virtual CC may be information indicating that the virtual CC including a plurality of BWPs and the virtual CC including a plurality of PRBs are supported.

[0062] Furthermore, the terminal 20 may assume that an index for identifying each virtual CC is configured by the base station 10 by RRC. Furthermore, the terminal 20 may assume that the index for identifying each virtual CC is a minimum value (for example, $i=0$ in FIG. 3 or 4) or a maximum value (for example, $i=1$ in FIG. 3 or 4) of the indexes of the component carriers.

[0063] The terminal 20 may assume that the unit of scheduling in the non-continuous scheduling is notified by (i) a virtual CC index, (ii) indexes of a plurality of component carriers+indexes of a plurality of BWPs, (iii) indexes of a plurality of component carriers+indexes of a plurality of PRBs or PRB sets, (iv) indexes of a plurality of component carriers+indexes of a plurality of BWPs+indexes of a plurality of PRBs or PRB sets, or the like.

[0064] Furthermore, the terminal 20 may assume that the unit of resources in carrier aggregation is virtual CCs, BWPs, PRBs, or PRB sets.

[0065] According to the above-described operation, resource allocation in the unit of scheduling having a granularity different from that of the component carrier can be realized.

[0066] Here, an operation related to an HARQ-ACK feedback has not been stipulated in the case where a virtual CC having a discontinuous actual frequency domain is introduced.

[0067] Therefore, the operation related to the HARQ-ACK feedback when the virtual CC is introduced may be executed as described below with respect to determination of a PUCCH cell, determination of an HARQ-ACK feedback timing, and construction of an HARQ-ACK codebook in the virtual CC.

[0068] It is noted that, in the embodiment of the present invention, a PUCCH is an example, and only needs to be a physical channel that transmits UL control information UCI. It may be replaced with a PUSCH. In addition, the UL control information may be at least one of SR, HARQ-ACK, or CSI, or may include other information. It is noted that the HARQ-ACK feedback may be replaced with SR or CSI. A type of the HARQ-ACK codebook may be different from a type to be described

below, or may be replaced with an existing type.

[0069] FIG. 5 is a diagram illustrating an example of a type-1 HARQ-ACK codebook. An HARQ-ACK bit is transmitted in a single PUCCH resource. The noted bit is referred to as an HARQ-ACK codebook. One of the two types is configured by an RRC parameter pdsch-HARQ-ACK-Codebook (refer to Non-Patent Literature 3).

[0070] FIG. 5 illustrates an example of the type-1 HARQ-ACK codebook, which is a semi-static codebook. The HARQ-ACK bit corresponds to PDSCH reception that may possibly be actually transmitted. In the example of FIG. 5, two types of values of 2 and 3 are set as $K_{sub.1}$ values, which are slot offsets of the PDSCH and the PUSCH, by RRC parameter dl-DataToUL-ACK (refer to Non-Patent Literature 3).

[0071] Resource allocation in the time domain is determined by RRC parameter pdsch-TimeDomainAllocationList (refer to Non-Patent Literature 3), as illustrated in FIG. 5. As illustrated in FIG. 5, the HARQ-ACK codebook including the HARQ-ACK bit for each of “A”, “B”, and “C” is constructed. It is noted that “A” corresponds to two cases, that is, a case in which the PDSCH is arranged from a symbol #2 to a symbol #14 and a case in which the PDSCH is arranged from a symbol #0 to a symbol #6 as illustrated in FIG. 5, and these cases are mutually exclusive.

[0072] FIG. 6 is a diagram illustrating an example of a type-2 HARQ-ACK codebook. FIG. 6 illustrates an example of the type-2 HARQ-ACK codebook, which is a dynamic codebook. The HARQ-ACK bit corresponds to PDSCH reception that is estimated to be actually transmitted.

[0073] The PDSCH reception can be estimated by a downlink assignment index (DAI). A counter DAI (C-DAI) indicates the number of PDSCH allocations up to the PDSCH. A total DAI (T-DAI) indicates the number of PDSCH allocations up to the slot. A remainder value of 4 is used for the DAI.

[0074] For example, when decoding of the DCI of “A” illustrated in FIG. 5 fails, since (C-DAI, T-DAI) of the preceding and subsequent DCI is (3, 1) and (1, 1), it can be determined that DCI of (0, 1) is missing, and NACK can be reported using the corresponding HARQ-ACK bit.

[0075] FIG. 7 is a diagram illustrating an example of a PUCCH group. PUCCH transmission for the PUCCH group for a plurality of CCs is supported by FR1 and FR2. In the same PUCCH group, up to four different SCSs are supported depending on a UE capability. Up to two PUCCH groups are supported depending on the UE capability. In the example of FIG. 7, a primary PUCCH group from CC #x1 to CC #m and a secondary PUCCH group from CC #y1 to CC #n are configured.

[0076] Regarding a cell of the PUCCH that transmits the HARQ-ACK feedback, determination may be made as follows:

[0077] The terminal 20 may transmit the PUCCH using a single actual CC included in the virtual CC. FIG. 8 is a diagram illustrating an example (1) of a PUCCH cell according to the embodiment of the present invention. As illustrated in FIG. 8, SCSs of the actual CCs constituting the virtual CC may all be the same. In the example of FIG. 8, all the PUCCHs are transmitted by using an actual CC #0. As illustrated in FIG. 8, all the PUCCHs may be transmitted by the same actual CC.

[0078] FIG. 9 is a diagram illustrating an example (2) of the PUCCH cell according to the embodiment of the present invention. As illustrated in FIG. 9, SCSs of the actual CCs constituting the virtual CC may be different from each other. In the example of FIG. 9, the PUCCH is transmitted by the actual CC #0 or an actual CC #1. Further, as illustrated in FIG. 9, all the PUCCHs may be transmitted by the same actual CC.

[0079] FIG. 10 is a diagram illustrating an example (3) of the PUCCH cell according to the embodiment of the present invention. As illustrated in FIG. 10, the PUCCH may be transmitted over a plurality of actual CCs included in the virtual CC. For example, in a virtual CC #0 illustrated in FIG. 10, the PUCCH is transmitted in the actual CC #0 and the actual CC #1, and in a virtual CC #1 illustrated in FIG. 10, the PUCCH is transmitted in an actual CC #2 and an actual CC #3. A constraint on a frequency resource of the PUCCH may be stipulated for each actual CC. For example, it may be specified that the frequency resource of the PUCCH is equal to or larger than

the number of RBs X or equal to or larger than the number of REs X for each actual CC.

[0080] As in the virtual CC #0 illustrated in FIG. 10, the SCSs of the actual CCs constituting the virtual CC may all be the same as each other. As in the virtual CC #1 illustrated in FIG. 11, the SCSs of the actual CCs constituting the virtual CC may be different from each other. In addition, the time resources for transmitting the PUCCH may have absolute times that are the same or different for the actual CCs. In addition, the time resources for transmitting the PUCCH may have the numbers of symbols that are the same or different for the actual CCs.

[0081] As illustrated in FIG. 10, when the PUCCH is transmitted over a plurality of actual CCs included in the virtual CC, a PUCCH format may be determined as 1) or 2) described below.

[0082] 1) When a PUCCH is transmitted over a plurality of actual CCs, a PUCCH format by sequence selection, such as a format 0 or a format 1 of an existing technical specification, does not need to be supported. [0083] 2) When the PUCCH is transmitted over the plurality of actual CCs, the PUCCH format may be determined depending on the size of a transmitted payload.

[0084] As illustrated in FIG. 10, when the PUCCH is transmitted over the plurality of actual CCs included in the virtual CC, the number of RBs may be determined as 1) or 2) described below.

[0085] 1) The number of RBs may be determined depending on a payload size, a modulation order, a coding rate, and the number of symbols to be transmitted, or whether to transmit the PUCCH over the plurality of actual CCs may be determined depending on the determined number of RBs.

[0086] 2) When a lower limit of the number of RBs is set for each actual CC, the number of transmission RBs in each actual CC may be determined based on the lower limit number of RBs.

[0087] FIG. 11 is a diagram illustrating an example (4) of the PUCCH cell according to the embodiment of the present invention. As illustrated in FIG. 11, a PUCCH group including a plurality of virtual CCs may be configured, and an HARQ-ACK feedback of a PDSCH scheduled to the plurality of virtual CCs may be aggregated and transmitted using the PUCCH in the single virtual CC. In the example of FIG. 11, the HARQ-ACK feedback of the PDSCH scheduled to the virtual CC #0 and the virtual CC #1 is transmitted using the PUCCH in the virtual CC #1.

[0088] Regarding the virtual CCs included in the PUCCH group, the same SCS may be used within each virtual CC, and different SCSs may be used for different virtual CCs. In addition, with respect to the actual CCs included in the virtual CCs included in the PUCCH group, the SCSs may all be the same or different for the actual CCs. Depending on a UE capability, whether SCSs are the same or not for supported CCs may vary, and UE capability signaling for reporting whether the SCSs are the same or not for the supported CCs may be stipulated.

[0089] FIG. 12 is a diagram illustrating an example (1) of an HARQ-ACK feedback timing according to the embodiment of the present invention. FIG. 13 is a diagram illustrating an example (2) of the HARQ-ACK feedback timing according to the embodiment of the present invention. As illustrated in FIGS. 12 and 13, an interval K.sub.1 between the PDSCH reception and the PUCCH that transmits the HARQ-ACK may be stipulated. The unit of K.sub.1 may be any one of a symbol, a slot, a slot group, or a millisecond. It is noted that, in the PDSCH, a resource may be allocated to a single actual CC, or one PDSCH resource may be allocated over a plurality of actual CCs. In addition, when the PDSCH is configured over a plurality of actual CCs, time domain resource allocation (TDRA) of the PDSCH may be the same or different for the actual CCs.

[0090] K.sub.1 may be any one of an option 1) to an option 6) described below. K.sub.1 in the option 1) to the option 4) illustrated in FIGS. 12 and 13 corresponds to the following description.

[0091] Option 1) An interval between a PDSCH (a slot or a symbol) and a PUCCH (a slot or a symbol) in an actual CC or a virtual CC that receives the PDSCH. The interval therebetween may be measured by the number of slots or symbols in the CC having the maximum or minimum SCS among the plurality of actual CCs to which the PDSCH is allocated. [0092] Option 2) An interval between a PDSCH (a slot or a symbol) and a PUCCH (a slot or a symbol) in an actual CC or a virtual CC that transmits the PUCCH. The interval therebetween may be measured by the number of slots or symbols in the CC having the maximum or minimum SCS among the plurality of actual

CCs to which the PUCCH is allocated. [0093] Option 3) An interval between the head (a slot or a symbol) of a PDSCH and a PUCCH (a slot or a symbol). [0094] Option 4) An interval between the tail end (a slot or a symbol) of a PDSCH and a PUCCH (a slot or a symbol). [0095] Option 5) An interval between a PDSCH (a slot or a symbol) and a PUCCH (a slot or a symbol) in an actual CC or a virtual CC that receives the PDCCH. [0096] Option 6) An interval between a PDSCH (a slot or a symbol) and a PUCCH (a slot or a symbol) in an actual CC having the minimum or maximum CC index among the actual CCs or virtual CCs for PDCCH transmission, PDSCH transmission, or PUCCH reception.

[0097] It is noted that a plurality of the options may be combined with each other. It is noted that the time resources for transmitting the PUCCH may have absolute times that are the same or different for the actual CCs, or may have the numbers of symbols that are the same or different.

[0098] The HARQ-ACK codebook may be stipulated as 1) to 3) described below. [0099] 1) The number of code words scheduled in one DCI may be set for each virtual CC or may be set for each actual cc. [0100] 2) Spatial bundling may be configured for each virtual CC or may be configured for each actual CC. [0101] 3) Code block group (CBG) based transmission may be configured for each virtual CC or may be configured for each actual CC.

[0102] Regarding a type-1 HARQ-ACK codebook, a codebook construction order of PDSCHs with the same PDSCH candidate occasion may have any one of the following sequences. It is noted that a PDSCH candidate opportunity may be a symbol, a slot, or a slot group in which the head or the tail end of the PDSCH can be configured.

[0103] FIG. 14 is a diagram illustrating an example (1) of the type-1 HARQ-ACK codebook according to the embodiment of the present invention. As illustrated in FIG. 14, HARQ-ACK bits may be generated in order from a CC having a smaller actual CC index to a CC having a larger actual CC index, and may further be generated in order from a CC having a smaller virtual CC index to a CC having a larger virtual CC index among the PDSCHs of the actual CC. That is, the DAI may be counted by giving the actual CC index the first priority and the virtual CC index the second priority. The example of FIG. 14 assumes a case in which one actual CC is included in a plurality of virtual CCs.

[0104] FIG. 15 is a diagram illustrating an example (2) of the type-1 HARQ-ACK codebook according to the embodiment of the present invention. As illustrated in FIG. 15, HARQ-ACK bits may be generate in order from a CC having a smaller virtual CC index to a CC having a larger virtual CC index, and may further be generated in order from a CC having a smaller actual CC index to a CC having a larger actual CC index among the PDSCHs of the virtual CC. That is, the DAI may be counted by giving the virtual CC index the first priority and the actual CC index the second priority. The example of FIG. 15 assumes a case in which a plurality of actual CCs are included in the single virtual CC.

[0105] FIG. 16 is a diagram illustrating an example (3) of the type-1 HARQ-ACK codebook according to the embodiment of the present invention. As illustrated in FIG. 16, HARQ-ACK bits may be generated in order from a CC having a smaller virtual CC index to a CC having a larger virtual CC index. Regarding a PDSCH reception candidate, a codebook may be generated assuming that PDSCHs having different actual CCs may be received simultaneously.

[0106] FIG. 17 is a diagram illustrating an example (4) of the type-1 HARQ-ACK codebook according to the embodiment of the present invention. FIG. 17 is an example of the type-1 HARQ-ACK codebook that is constructed by being arranged first by the virtual CC indexes and then by the actual CC indexes as illustrated in FIG. 15. As illustrated in FIG. 17, the type-1 HARQ-ACK codebook is constructed in the following order: a PDSCH #1 of the actual CC #0, a PDSCH #2 of the actual CC #2, a PDSCH #3 of the actual CC #1, a PDSCH #4 of the actual CC #3, a PDSCH #5 of the actual CC #0, a PDSCH #6 of the actual CC #2, a PDSCH #7 of the actual CC #1, and a PDSCH #8 of the actual CC #3.

[0107] FIG. 18 is a diagram illustrating an example (5) of the type-1 HARQ-ACK codebook

according to the embodiment of the present invention. FIG. 18 is an example of the type-1 HARQ-ACK codebook that is constructed by being arranged first by the actual CC indexes and then by the virtual CC indexes as illustrated in FIG. 14. As illustrated in FIG. 14, the type-1 HARQ-ACK codebook is constructed in the following order: the PDSCH #1 of the actual CC #0, the PDSCH #2 of the actual CC #1, the PDSCH #3 of the actual CC #2, the PDSCH #4 of the actual CC #3, the PDSCH #5 of the actual CC #0, the PDSCH #6 of the actual CC #2, the PDSCH #7 of the actual CC #1, and the PDSCH #8 of the actual CC #3.

[0108] Regarding the type-2 HARQ-ACK codebook, DAI counting may be performed as in 1) or 2) described below. [0109] 1) The DAI may be counted by DCI. When the DCI for scheduling one PDSCH has resources allocated over a plurality of actual CCs, the DAI may be counted for each actual CC as one DCI, or the DAI may be counted as one DCI. [0110] 2) The DAI may be counted using the PDSCH. When one PDSCH has resources allocated over a plurality of actual CCs, the DAI may be counted for each actual CC as one DCI, or the DAI may be counted as one DCI.

[0111] The type-2 HARQ-ACK codebook may be constructed as in an option A) or an option B).

[0112] Option A) Sub-codebooks may be constructed for each one of the virtual CCs and concatenated. [0113] Option B) A codebook may be constructed using the virtual CCs altogether.

Further, DAI counting may be performed as in 1) or 2) described below. [0114] 1) HARQ-ACK bits may be generated in the order from a CC having a smaller actual CC index to a CC having a larger actual CC index, and may be further generated in the order from a CC having a smaller virtual CC index to a CC having a larger virtual CC index among the DCI or the PDSCHs of the actual CC. That is, the DAI may be counted by giving the actual CC index the first priority and the virtual CC index the second priority. It is assumed that one actual CC is included in a plurality of virtual CCs. [0115] 2) HARQ-ACK bits may be generated in the order from a CC having a smaller virtual CC index to a CC having a larger virtual CC index, and may be further generated in the order from a CC having a smaller actual CC index to a CC having a larger actual CC index among the DCI or the PDSCHs of the virtual CC. That is, the DAI may be counted by giving the virtual CC index the first priority and the actual CC index the second priority. It is assumed that a plurality of actual CCs are included in a single virtual CC.

[0116] Regarding ACK/NACK bits of the HARQ-ACK, when one PDSCH is scheduled over a plurality of actual CCs, the ACK/NACK bits may be aggregated, or a different ACK/NACK bit may be provided for each actual CC.

[0117] FIG. 19 is a diagram illustrating an example (1) of the type-2 HARQ-ACK codebook according to the embodiment of the present invention. FIG. 19 is an example of codebook construction for the option A) and illustrates an example in which the PDSCH is scheduled by being confined within a single actual CC. Furthermore, FIG. 19 is an example of counting the DAI with respect to the DCI.

[0118] As illustrated in FIG. 19, the ACK/NACK bits of the 1st sub-codebook for the virtual CC #0 are such that (C-DAI, T-DAI) is in the following order: (0, 1) of the actual CC #0, (1, 1) of the actual CC #2, (2, 3) of the actual CC #0, and (3, 3) of the actual CC #2.

[0119] As illustrated in FIG. 19, the ACK/NACK bits of the 2nd sub-codebook for the virtual CC #1 are such that (C-DAI, T-DAI) is in the following order: (0, 1) of the actual CC #1, (1, 1) of the actual CC #3, (2, 3) of the actual CC #1, and (3, 3) of the actual CC #3.

[0120] FIG. 20 is a diagram illustrating an example (2) of the type-2 HARQ-ACK codebook according to the embodiment of the present invention. FIG. 20 is an example of codebook construction for the option B) and illustrates an example in which the PDSCH is scheduled over a plurality of actual CCs. Furthermore, FIG. 20 is an example of counting the DAI with respect to the DCI. Additionally, FIG. 20 is an example of aggregating the ACK/NACK bits of the PDSCH scheduled over a plurality of actual CCs. Further, FIG. 20 is an example of counting the DAI for each virtual CC index.

[0121] As illustrated in FIG. 20, the ACK/NACK bits of the codebook are such that (C-DAI, T-

DAI) is in the following order: (0, 1) of the actual CC #0 and the actual CC #2, (1, 1) of the actual CC #1 and the actual CC #3, (2, 3) of the actual CC #0 and the actual CC #2, and (3, 3) of the actual CC #1 and the actual CC #3.

[0122] FIG. 21 is a diagram illustrating an example (3) of the type-2 HARQ-ACK codebook according to the embodiment of the present invention. FIG. 21 is an example of codebook construction for the option B) and illustrates an example in which the PDSCH is scheduled by being confined within a single actual CC. Furthermore, FIG. 21 is an example of counting the DAI with respect to the DCI. Additionally, FIG. 21 is an example of counting the DAI by prioritizing the virtual CC index in the first order and the actual CC index in the second order.

[0123] As illustrated in FIG. 21, the ACK/NACK bits are such that (C-DAI, T-DAI) is in the following order: (0, 3) of the actual CC #0, (1, 3) of the actual CC #2, (2, 3) of the actual CC #1, (3, 3) of the actual CC #3, (0, 1) of the actual CC #0, (1, 1) of the actual CC #2, (2, 3) of the actual CC #1, and (3, 3) of the actual CC #3.

[0124] FIG. 22 is a diagram illustrating an example (4) of the type-2 HARQ-ACK codebook according to the embodiment of the present invention. FIG. 22 is an example of codebook construction for the option B) and illustrates an example in which the PDSCH is scheduled over a plurality of actual CCs. Furthermore, FIG. 22 is an example of counting the DAI with respect to the PDSCH. Additionally, FIG. 22 is an example of counting the DAI by giving the actual CC index the first priority and the virtual CC index the second priority.

[0125] As illustrated in FIG. 21, the ACK/NACK bits are such that (C-DAI, T-DAI) is in the following order: (0, 3) of the actual CC #0, (1, 3) of the actual CC #1, (2, 3) of the actual CC #2, (3, 3) of the actual CC #3, (0, 1) of the actual CC #0, (1, 1) of the actual CC #2, (2, 3) of the actual CC #1, and (3, 3) of the actual CC #3.

[0126] According to the above-described embodiment, when a virtual CC discontinuous in the frequency domain is introduced, the base station 10 and the terminal 20 can determine the timing for transmitting a PUCCH and an HARQ-ACK codebook, and can execute an HARQ-ACK feedback.

[0127] That is, in a wireless communication system, a feedback operation related to retransmission control can be executed in a wideband that is discontinuous in the frequency domain.

Device Configuration

[0128] In the following, a description will be given as to functional example configurations of the base station 10 and the terminal 20 that execute processes and functions described above. The base station 10 and the terminal 20 include the function to implement the above-described embodiment. However, each of the base station 10 and the terminal 20 may have only some of the functions of the embodiment.

<Base Station 10>

[0129] FIG. 23 is a diagram illustrating an example of the functional configuration of the base station 10 according to the embodiment of the present invention. As illustrated in FIG. 23, the base station 10 includes a transmitting unit 110, a receiving unit 120, a configuring unit 130, and a control unit 140. The functional configuration illustrated in FIG. 23 is merely an example. As long as the functions of the embodiment of the present invention can be executed, the functional classification and the names of functional units may be determined freely.

[0130] The transmitting unit 110 has the function to generate signals to be transmitted to the terminal 20 side and wirelessly transmitting the signals. In addition, the transmitting unit 110 transmits an inter-network node message to another network node. The receiving unit 120 has the function to receive various signals transmitted from the terminal 20 and acquire, for example, information of a higher layer from the received signals. In addition, the transmitting unit 110 has the function to transmit NR-PSS, NR-SSS, NR-PBCH, DL/UL control signals, and the like to the terminal 20. Further, the receiving unit 120 receives the inter-network node message from another network node.

[0131] The configuring unit **130** stores configuration information configured in advance and various types of configuration information to be transmitted to the terminal **20**. One of the contents of the configuration information is, for example, information related to multi-carrier scheduling.

[0132] As described in the embodiment, the control unit **140** performs control related to the multi-carrier scheduling. A functional unit related to signal transmission in the control unit **140** may be included in the transmitting unit **110**, and a functional unit related to signal reception in the control unit **140** may be included in the receiving unit **120**.

<Terminal **20**>

[0133] FIG. **24** is a diagram illustrating an example of the functional configuration of the terminal **20** according to the embodiment of the present invention. As illustrated in FIG. **24**, the terminal **20** includes a transmitting unit **210**, a receiving unit **220**, a configuring unit **230**, and a control unit **240**. The functional configuration illustrated in FIG. **24** is merely an example. As long as the functions of the embodiment of the present invention can be executed, the functional classification and the names of functional units may be determined freely.

[0134] The transmitting unit **210** generates transmission signals from transmission data and wirelessly transmits the transmission signals. The receiving unit **220** wirelessly receives various signals and acquires higher layer signals from the received physical layer signals. In addition, the receiving unit **220** has the function to receive NR-PSS, NR-SSS, NR-PBCH, DL/UL/SL control signals, and the like transmitted from the base station **10**. Furthermore, for example, the transmitting unit **210** transmits a physical sidelink control channel (PSCCH), a physical sidelink shared channel (PSSCH), a physical sidelink discovery channel (PSDCH), a physical sidelink broadcast channel (PSBCH), or the like to the other terminal **20** as D2D communication, and the receiving unit **220** receives the PSCCH, the PSSCH, the PSDCH, the PSBCH, or the like from the other terminal **20**.

[0135] The configuring unit **230** stores various types of configuration information received from the base station **10** by the receiving unit **220**. The configuring unit **230** also stores configuration information configured in advance. One of the contents of the configuration information is, for example, information related to multi-carrier scheduling.

[0136] As described in the embodiment, the control unit **240** performs control related to the multi-carrier scheduling. A functional unit related to signal transmission in the control unit **240** may be included in the transmitting unit **210**, and a functional unit related to signal reception in the control unit **240** may be included in the receiving unit **220**.

Hardware Configuration

[0137] The block diagrams (FIGS. **23** and **24**) used for the description of the above-described embodiment illustrate blocks of functional units. These functional blocks (configuration units) are realized by any combination of at least one of hardware or software. A method of implementing each functional block is not particularly limited. That is, each functional block may be realized by using one physically or logically combined device, or may be realized by directly or indirectly (for example, by using wired, wireless, or the like) connecting two or more physically or logically separated devices and using a plurality of these devices. The functional block may be realized by combining software with the one device or the plurality of devices.

[0138] The functions include, but are not limited to, determining, deciding, judging, calculating, computing, processing, deriving, investigating, searching, ascertaining, receiving, transmitting, outputting, accessing, resolving, selecting, choosing, establishing, comparing, assuming, expecting, considering, broadcasting, notifying, communicating, forwarding, configuring, reconfiguring, allocating, mapping, assigning, and the like. For example, a functional block (a configuration unit) that causes transmission to function is referred to as a transmitting unit or a transmitter. In any case, as described above, the implementation method is not particularly limited.

[0139] For example, the base station **10**, the terminal **20**, and the like in the embodiment of the present disclosure may function as a computer that performs processing of the wireless

communication method of the present disclosure. FIG. 25 is a diagram illustrating an example of the hardware configuration of the base station **10** and the terminal **20** according to the embodiment of the present disclosure. The base station **10** and the terminal **20** described above may be physically configured as a computer device including a processor **1001**, a storage device **1002**, an auxiliary storage device **1003**, a communication device **1004**, an input device **1005**, an output device **1006**, a bus **1007**, and the like.

[0140] In the following description, the term “device” can be replaced with a circuit, a device, a unit, or the like. The hardware configuration of the base station **10** and the terminal **20** may be configured to include one or a plurality of devices illustrated in the drawing, or may be configured without including some devices.

[0141] Each function in the base station **10** and the terminal **20** is implemented by the processor **1001** performing arithmetic operations upon loading predetermined software (a program) to hardware such as the processor **1001** and the storage device **1002**, thereby controlling communication by the communication device **1004**, and controlling at least one of reading or writing of data in the storage device **1002** and the auxiliary storage device **1003**.

[0142] The processor **1001** runs, for example, an operating system to perform overall control of the computer. The processor **1001** may be configured with a central processing unit (CPU) including an interface with a peripheral device, a control device, an arithmetic device, a register, and the like. For example, the control unit **140**, the control unit **240**, and the like described above may be realized by the processor **1001**.

[0143] In addition, the processor **1001** loads a program (a program code), a software module, data, or the like from at least one of the auxiliary storage device **1003** or the communication device **1004** to the storage device **1002**, and executes various types of processing according to the data. As the program, a program that causes a computer to execute at least a part of the operations described in the above-described embodiments is used. For example, the control unit **140** of the base station **10** illustrated in FIG. 23 may be realized by a control program stored in the storage device **1002** and operated by the processor **1001**. Furthermore, for example, the control unit **240** of the terminal **20** illustrated in FIG. 24 may be realized by a control program stored in the storage device **1002** and operated by the processor **1001**. Although it has been described that the above-described various types of processing are executed by one processor **1001**, the various types of processing may be executed simultaneously or sequentially by two or more processors **1001**. The processor **1001** may be mounted by one or more chips. It is noted that the program may be transmitted from a network via an electric communication line.

[0144] The storage device **1002** is a computer-readable recording medium, and may be configured with, for example, at least one of a read only memory (ROM), an erasable programmable ROM (EPROM), an electrically erasable programmable ROM (EEPROM), a random access memory (RAM), or the like. The storage device **1002** may be referred to as a register, a cache, a main memory (a main storage device), or the like. The storage device **1002** can store a program (a program code), a software module, and the like that can be executed to implement the communication method according to the embodiment of the present disclosure.

[0145] The auxiliary storage device **1003** is a computer-readable recording medium, and may be configured with, for example, at least one of an optical disk such as a compact disc ROM (CD-ROM), a hard disk drive, a flexible disk, a magneto-optical disk (for example, a compact disc, a digital versatile disc, or a Blu-ray (registered trademark) disc), a smart card, a flash memory (for example, a card, a stick, or a key drive), a floppy (registered trademark) disk, a magnetic strip, or the like. The above-described storage medium may be, for example, a database including at least one of the storage device **1002** or the auxiliary storage device **1003**, a server, or another appropriate medium.

[0146] The communication device **1004** is hardware (a transmitting and receiving device) for performing communication between computers via at least one of a wired network or a wireless

network, and is also referred to as, for example, a network device, a network controller, a network card, a communication module, or the like. The communication device **1004** may be configured with a high-frequency switch, a duplexer, a filter, a frequency synthesizer, and the like in order to implement, for example, at least one of frequency division duplex (FDD) and time division duplex (TDD). For example, a transmitting and receiving antenna, an amplifier unit, a transmitting and receiving unit, a transfer path interface, and the like may be realized by the communication device **1004**. The transmitting and receiving unit may be implemented as physically or logically separated transmitting unit and the receiving unit.

[0147] The input device **1005** is an input device (for example, a keyboard, a mouse, a microphone, a switch, a button, a sensor, or the like) that receives an input from the outside. The output device **1006** is an output device (for example, a display, a speaker, an LED lamp, and the like) that executes an output to the outside. It is noted that the input device **1005** and the output device **1006** may be formed to be integrated with each other (for example, a touch panel).

[0148] In addition, the respective devices such as the processor **1001** and the storage device **1002** are connected to each other by the bus **1007** for communicating information. The bus **1007** may be configured using a single bus or may be configured using different buses between the devices.

[0149] Furthermore, the base station **10** and the terminal **20** may be configured with hardware such as a microprocessor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a programmable logic device (PLD), or a field programmable gate array (FPGA), and some or all of the functional blocks may be realized by the hardware. For example, the processor **1001** may be mounted using at least one of these pieces of hardware.

[0150] FIG. **26** illustrates an example configuration of a vehicle **2001**. As illustrated in FIG. **26**, the vehicle **2001** includes a drive unit **2002**, a steering unit **2003**, an accelerator pedal **2004**, a brake pedal **2005**, a shift lever **2006**, a front wheel **2007**, a rear wheel **2008**, an axle **2009**, an electronic control unit **2010**, various sensors **2021** to **2029**, an information service unit **2012**, and a communication module **2013**. Each aspect/embodiment described in the present disclosure may be applied to a communication device mounted on the vehicle **2001**, and for example, may be applied to the communication module **2013**.

[0151] The drive unit **2002** includes, for example, an engine, a motor, or a hybrid of an engine and a motor. The steering unit **2003** includes at least a steering wheel (also referred to as a driving wheel), and is configured to steer at least one of the front wheel or the rear wheel based on an operation of the steering wheel operated by a user.

[0152] The electronic control unit **2010** includes a microprocessor **2031**, a memory (ROM, RAM) **2032**, and a communication port (IO port) **2033**. Signals from the various sensors **2021** to **2029** provided in the vehicle **2001** are input to the electronic control unit **2010**. The electronic control unit **2010** may be referred to as an electronic control unit (ECU).

[0153] Examples of the signals from the various sensors **2021** to **2029** include a current signal from the current sensor **2021** that senses a current of a motor, a rotation speed signal of a front wheel and a rear wheel acquired by the rotation speed sensor **2022**, an air pressure signal of a front wheel and a rear wheel acquired by the air pressure sensor **2023**, a vehicle speed signal acquired by the vehicle speed sensor **2024**, an acceleration signal acquired by the acceleration sensor **2025**, a depression amount signal of an accelerator pedal acquired by the accelerator pedal sensor **2029**, a depression amount signal of a brake pedal acquired by the brake pedal sensor **2026**, an operation signal of a shift lever acquired by the shift lever sensor **2027**, and a detection signal for detecting an obstacle, a vehicle, a pedestrian, and the like acquired by the object detection sensor **2028**.

[0154] The information service unit **2012** includes various devices for providing (outputting) various types of information such as driving information, traffic information, and entertainment information, such as a car navigation system, an audio system, a speaker, a television, and a radio, and one or more ECUs that control these devices. The information service unit **2012** provides various kinds of multi-media information and multi-media services to an occupant in the vehicle

2001 using information acquired from an external device via the communication module **2013** or the like. The information service unit **2012** may include an input device (for example, a keyboard, a mouse, a microphone, a switch, a button, a sensor, a touch panel, or the like) that receives an input from the outside, or may include an output device (for example, a display, a speaker, an LED lamp, a touch panel, or the like) that executes an output to the outside.

[0155] A driving assistance system unit **2030** includes various devices for providing functions of preventing an accident in advance and reducing a driver's driving load, such as a millimeter wave radar, light detection and ranging (LiDAR), a camera, a positioning locator (for example, GNSS or the like), map information (for example, a high definition (HD) map, an autonomous vehicle (AV) map, and the like), a gyro system (for example, an inertial measurement unit (IMU), an inertial navigation system (INS), or the like), an artificial intelligence (AI) chip, and an AI processor, and one or more ECUs for controlling these devices. The driving assistance system unit **2030** also transmits and receives various types of information via the communication module **2013**, thereby achieving a driving assistance function or an automatic driving function.

[0156] The communication module **2013** may communicate with the microprocessor **2031** and components of the vehicle **2001** via a communication port. For example, the communication module **2013** transmits and receives, via the communication port **2033**, data to and from the drive unit **2002**, the steering unit **2003**, the accelerator pedal **2004**, the brake pedal **2005**, the shift lever **2006**, the front wheel **2007**, the rear wheel **2008**, the axle **2009**, the microprocessor **2031** and the memory (ROM, RAM) **2032** in the electronic control unit **2010**, and the sensors **2021** to **2029** provided in the vehicle **2001**.

[0157] The communication module **2013** is a communication device that can be controlled by the microprocessor **2031** of the electronic control unit **2010** and can communicate with an external device. For example, various types of information are transmitted and received to and from an external device via wireless communication. The communication module **2013** may be provided in the inside or outside of the electronic control unit **2010**. The external device may be, for example, a base station, a mobile station, or the like.

[0158] The communication module **2013** may transmit at least one of the above-described signals from the various sensors **2021** to **2028** input to the electronic control unit **2010**, information obtained based on the signals, and information based on an input from the outside (user) obtained via the information service unit **2012** to the external device via wireless communication. The electronic control unit **2010**, the various sensors **2021** to **2028**, the information service unit **2012**, and the like may be referred to as input units that receive inputs. For example, the PUSCH transmitted by the communication module **2013** may include information based on the input.

[0159] The communication module **2013** receives various types of information (traffic information, signal information, inter-vehicle information, and the like) transmitted from the external device, and displays the information on the information service unit **2012** provided in the vehicle **2001**. The information service unit **2012** may be referred to as an output unit that outputs information (for example, information is output to a device such as a display or a speaker based on a PDSCH (or data/information decoded from the PDSCH) received by the communication module **2013**). The communication module **2013** also stores various types of information received from the external devices in the memory **2032** available by the microprocessor **2031**. The microprocessor **2031** may control, based on the information stored in the memory **2032**, the drive unit **2002**, the steering unit **2003**, the accelerator pedal **2004**, the brake pedal **2005**, the shift lever **2006**, the front wheel **2007**, the rear wheel **2008**, the axle **2009**, the sensors **2021** to **2029**, and the like included in the vehicle **2001**.

Summary of Embodiments

[0160] As described above, according to the embodiment of the present invention, there is provided a terminal including a receiving unit configured to receive, from a base station, a downlink shared channel in a virtual component carrier (CC) that is discontinuous in a frequency domain and that

includes a plurality of actual CCs, a control unit configured to determine feedback information related to retransmission control corresponding to the downlink shared channel, and a transmitting unit configured to transmit, to the base station, an uplink control channel including the feedback information in the virtual CC, wherein the control unit determines which actual CC included in the virtual CC transmits the uplink control channel, and determines an interval between the downlink shared channel and the uplink control channel.

[0161] According to the configuration, when a virtual CC having a discontinuous frequency domain is introduced, the base station **10** and the terminal **20** can determine the timing for transmitting a PUCCH and an HARQ-ACK codebook, and can execute an HARQ-ACK feedback. That is, in a wireless communication system, a feedback operation related to retransmission control can be executed in a wideband that is discontinuous the frequency domain.

[0162] The control unit may determine the interval between the downlink shared channel and the uplink control channel in an actual CC receiving the downlink shared channel. According to the configuration, when a virtual CC having a discontinuous frequency domain is introduced, the base station **10** and the terminal **20** can determine the timing for transmitting a PUCCH and an HARQ-ACK codebook, and can execute an HARQ-ACK feedback.

[0163] The control unit may determine the interval between the downlink shared channel and the uplink control channel in an actual CC transmitting the uplink control channel. According to the configuration, when a virtual CC having a discontinuous frequency domain is introduced, the base station **10** and the terminal **20** can determine the timing for transmitting a PUCCH and an HARQ-ACK codebook, and can execute an HARQ-ACK feedback.

[0164] The control unit may determine an interval between a head of the downlink shared channel and the uplink control channel. According to the configuration, when a virtual having a discontinuous frequency domain is introduced, the base station **10** and the terminal **20** can determine the timing for transmitting a PUCCH and an HARQ-ACK codebook, and can execute an HARQ-ACK feedback.

[0165] The control unit may determine an interval between a tail end of the downlink shared channel and the uplink control channel. According to the configuration, when a virtual CC having a discontinuous frequency domain is introduced, the base station **10** and the terminal **20** can determine the timing for transmitting a PUCCH and an HARQ-ACK codebook, and can execute an HARQ-ACK feedback.

[0166] Furthermore, according to the embodiment of the present invention, there is provided a communication method to be performed by a terminal, including a reception procedure of receiving, from a base station, a downlink shared channel in a virtual component carrier (CC) that is discontinuous in a frequency domain and that includes a plurality of actual CCs, a control procedure of determining feedback information related to retransmission control corresponding to the downlink shared channel, a transmission procedure of transmitting, to the base station, an uplink control channel including the feedback information in the virtual CC, and a procedure of determining which actual CC included in the virtual CC transmits the uplink control channel and determining an interval between the downlink shared channel and the uplink control channel.

[0167] According to the configuration, when a virtual CC having a discontinuous frequency domain is introduced, the base station **10** and the terminal **20** can determine the timing for transmitting a PUCCH and an HARQ-ACK codebook, and can execute an HARQ-ACK feedback. That is, in a wireless communication system, a feedback operation related to retransmission control can be executed in a wideband that is discontinuous in the frequency domain.

Supplement to Embodiment

[0168] Although the embodiments of the present invention have been described above, the disclosed invention is not limited to such embodiments, and those skilled in the art will understand various modifications, modifications, alternatives, substitutions, and the like. Although the description has been given using specific numerical examples to facilitate understanding of the

invention, these numerical values are merely examples, and any appropriate value may be used, unless otherwise specified. The classification of items in the above description is not essential to the present invention, and items described in two or more items may be used in combination as necessary, or items described in one item may be applied to items described in another item (as long as there is no contradiction). A boundary of a functional unit or a processing unit in the functional block diagram does not necessarily correspond to a boundary of a physical component. The operation of the plurality of functional units may be physically performed by one component, or the operation of one functional unit may be physically performed by a plurality of components. Regarding a processing procedure described in the embodiments, the order of the processing may be changed as long as there is no contradiction. For convenience of processing description, the base station **10** and the terminal **20** have been described using a functional block diagram, but such a device may be realized by hardware, software, or a combination thereof. The software operated by the processor included in the base station **10** according to the embodiments of the present invention and the software operated by the processor included in the terminal **20** according to the embodiments of the present invention may be stored in any appropriate storage medium such as a random access memory (RAM), a flash memory, a read-only memory (ROM), an EPROM, an EEPROM, a register, a hard disk (HDD), a removable disk, a CD-ROM, a database, or a server.

[0169] Furthermore, the notification of information is not limited to the aspects/embodiments described in the present disclosure, and may be performed using other methods. For example, the notification of information may be performed by physical layer signaling (for example, downlink control information (DCI) and uplink control information (UCI)), higher layer signaling (for example, radio resource control (RRC) signaling and medium access control (MAC) signaling), broadcast information (master information block (MIB) and system information block (SIB)), other signals, or a combination thereof. Furthermore, the RRC signaling may be referred to as an RRC message, and may be, for example, an RRC connection setup message, an RRC connection reconfiguration message, or the like.

[0170] Each aspect/embodiment described in the present disclosure may be applied to at least one of a system utilizing long term evolution (LTE), LTE-Advanced (LTE-A), SUPER 3G, IMT-Advanced, 4th generation mobile communication system (4G), 5th generation mobile communication system (5G), 6th generation mobile communication system (6G), xth generation mobile communication system (xG) (xG (x is, for example, an integer or a decimal)), future radio access (FRA), new radio (NR), new radio access (NX), future generation radio access (FX), W-CDMA (registered trademark), GSM (registered trademark), CDMA2000, ultra mobile wideband (UMB), IEEE 802.11 (Wi-Fi (registered trademark)), IEEE 802.16 (WiMAX (registered trademark)), IEEE 802.20, Ultra-WideBand (UWB), Bluetooth (registered trademark), other appropriate systems, or a next generation system that is extended, modified, generated, and stipulated based thereon. Further, a plurality of systems may be applied in combination (for example, a combination of at least one of LTE and LTE-A and 5G, and the like).

[0171] The order of the processing procedure, sequence, flowchart, and the like of each aspect/embodiment described in the present specification may be changed as long as there is no contradiction. For example, for the method described in the present disclosure, elements of various steps are presented using an example order, and are not limited to the presented particular order.

[0172] Specific operations described as being performed by the base station **10** in the present specification may be performed by an upper node thereof in some cases. In a network including one or a plurality of network nodes having the base station **10**, it is obvious that various operations performed for communication with the terminal **20** may be performed by at least one of the base station **10** or other network nodes (for example, MME, S-GW, or the like is conceivable, but the present invention is not limited thereto) other than the base station **10**. Although a description has been given as to a case in which there is one other network node other than the base station **10**, the other network node may be a combination of a plurality of other network nodes (for example,

MME and S-GW).

[0173] Information, a signal, or the like described in the present disclosure can be output from an upper layer (or a lower layer) to a lower layer (or an upper layer). Input and output may be performed via a plurality of network nodes.

[0174] The input/output information and the like may be stored in a specific location (for example, in a memory) or may be managed using a management table. The input/output information and the like can be overwritten, updated, or additionally written. The output information and the like may be deleted. The input information and the like may be transmitted to another device.

[0175] The determination in the present disclosure may be made by a value represented by one bit (0 or 1), may be made by a true/false value (Boolean: true or false), or may be made by comparison of numerical values (for example, comparison with a predetermined value).

[0176] Software, whether referred to as software, firmware, middleware, microcode, hardware description language, or other names, should be construed broadly to mean an instruction, an instruction set, a code, a code segment, a program code, a program, a subprogram, a software module, an application, a software application, a software package, a routine, a subroutine, an object, an executable file, an execution thread, a procedure, a function, and the like.

[0177] In addition, software, instructions, information, and the like may be transmitted and received via a transmission medium. For example, when software is transmitted from a website, a server, or other remote sources using at least one of a wired technology (a coaxial cable, an optical fiber cable, a twisted pair, a digital subscriber line (DSL), or the like) or a wireless technology (infrared rays, microwaves, and the like), at least one of these wired and wireless technologies is included within the definition of the transmission medium.

[0178] The pieces of information, signals, and the like described in the present disclosure may be represented using any one of a variety of different techniques. For example, data, an instruction, a command, information, a signal, a bit, a symbol, a chip, and the like that may be mentioned throughout the above description may be represented by a voltage, a current, an electromagnetic wave, a magnetic field or a particle, an optical field or a photon, or any combination thereof.

[0179] It is noted that the terms described in the present disclosure and the terms necessary for understanding the present disclosure may be replaced with terms having the same or similar meanings. For example, at least one of the channel or the symbol may be a signal (signaling). The signal may also be a message. In addition, a component carrier (CC) may be referred to as a carrier frequency, a cell, a frequency carrier, or the like.

[0180] The terms “system” and “network” used in the present disclosure are used interchangeably.

[0181] In addition, the information, the parameter, and the like described in the present disclosure may be represented using an absolute value, may be represented using a relative value from a predetermined value, or may be represented using another corresponding information. For example, a radio resource may be indicated by an index.

[0182] The names used for the parameters described above are not limited in any respect.

Furthermore, mathematical formulas and the like using these parameters may be different from those explicitly disclosed in the present disclosure. Since various channels (for example, PUCCH, PDCCH, and the like) and information elements can be identified by any suitable name, various names assigned to these various channels and information elements are not in any way limitative names.

[0183] In the present disclosure, terms such as “base station (BS)”, “radio base station”, “base station device”, “fixed station”, “NodeB”, “eNodeB (eNB)”, “gNodeB (gNB)”, “access point”, “transmission point”, “reception point”, “transmission/reception point”, “cell”, “sector”, “cell group”, “carrier”, and “component carrier” can be used interchangeably. The base station may also be referred to as a macro cell, a small cell, a femtocell, a picocell, or the like.

[0184] The base station may accommodate one or a plurality of (for example, three) cells. When the base station accommodates a plurality of cells, an entire coverage area of the base station may

be divided into a plurality of smaller areas, and each smaller area may also provide a communication service by a base station subsystem (for example, a small base station for indoor use (remote radio head (RRH))). The term “cell” or “sector” refers to a part or the whole of a coverage area of at least one of the base station or the base station subsystem that performs a communication service in this coverage.

[0185] In the present disclosure, a state indicating that the base station transmits information to the terminal may be replaced with a state indicating that the base station instructs the terminal to perform control and operation based on the information.

[0186] In the present disclosure, terms such as “mobile station (MS)”, “user terminal”, “user equipment (UE)”, and “terminal” can be used interchangeably.

[0187] The mobile station may also be referred to, by those skilled in the art, as a subscriber station, a mobile unit, a subscriber unit, a wireless unit, a remote unit, a mobile device, a wireless device, a wireless communications device, a remote device, a mobile subscriber station, an access terminal, a mobile terminal, a wireless terminal, a remote terminal, a handset, a user agent, a mobile client, a client, or some other suitable term.

[0188] At least one of the base station or the mobile station may be referred to as a transmitting device, a receiving device, a communication device, or the like. It is noted that at least one of the base station or the mobile station may be a device mounted on a mobile object, the mobile object itself, or the like. The mobile object refers to a movable object, and a moving speed thereof is any moving speed. In addition, a case in which the mobile object is stopped is naturally included. The mobile object includes, for example, a vehicle, a transport vehicle, an automobile, a motorcycle, a bicycle, a connected car, an excavator, a bulldozer, a wheel loader, a dump truck, a forklift, a train, a bus, a rear car, a human-powered vehicle, a ship and other watercraft, an airplane, a rocket, an artificial satellite, a drone (registered trademark), a multi-copter, a quadcopter, a balloon, and objects mounted thereon, and the present invention is not limited thereto. Further, the mobile object may be a mobile object that autonomously travels based on an operation command. The mobile object may be a vehicle (for example, a car, an airplane, or the like), a mobile object that is movable without human intervention (for example, a drone, a self-driving vehicle, or the like), or a robot (manned type or unmanned type). It is noted that at least one of the base station or the mobile station includes a device that does not necessarily move during the communication operation. For example, at least one of the base station or the mobile station may be an Internet of Things (IoT) device such as a sensor.

[0189] In addition, the base station in the present disclosure may be replaced with the user terminal. For example, each aspect/embodiment of the present disclosure may be applied to a configuration in which communication between a base station and the user terminal is replaced with communication between a plurality of terminals **20** (for example, may be referred to as device-to-device (D2D), vehicle-to-everything (V2X), or the like). In this case, the terminal **20** may have the same functions as those of the base station **10** described above. In addition, words such as “upstream” and “downstream” may be replaced with words corresponding to inter-terminal communication (for example, “side”). For example, an uplink channel, a downlink channel, and the like may be replaced with a side channel.

[0190] Similarly, the user terminal in the present disclosure may be replaced with the base station. In this case, the base station may have the same functions as those of the user terminal described above.

[0191] The terms “determining” and “determining” used in the present disclosure may encompass a wide variety of actions. The terms “determining” and “determining” may include, for example, a case in which “judging”, “calculating”, “computing”, “processing”, “deriving”, “investigating”, “looking up, searching, inquiring” (for example, looking up in a table, a database, or another data structure), and “ascertaining” is deemed as “determining” and “determining”. Furthermore, the terms “determining” and “determining” may include a case in which “receiving” (for example,

receiving information), “transmitting” (for example, transmitting information), “inputting”, “outputting”, or “accessing” (for example, accessing data in a memory) is deemed as “determining” and “determining”. In addition, the terms “determining” and “determining” may include a case in which “resolving”, “selecting”, “choosing”, “establishing”, “comparing”, or the like is deemed as “determining” and “determining”. That is, the “determining” and “determining” may include a case in which a certain action or operation is deemed as “determining” and “determining”. Further, “determining” may be replaced with “assuming”, “expecting”, “considering”, or the like.

[0192] The terms “connected”, “coupled”, or any variation thereof, mean any direct or indirect connection or coupling between two or more elements, and can include the presence of one or more intermediate elements between two elements that are “connected” to each other or “coupled” to each other. The coupling or connection between the elements may be physical, logical, or a combination thereof. For example, “connection” may be replaced with “access”. As used in this disclosure, two elements can be considered to be “connected” or “coupled” to each other using at least one of one or more wires, cables, or printed electrical connections, and also using, as some non-limiting and non-inclusive examples, electromagnetic energy having wavelengths in a radio frequency domain, a microwave region, and a light (both visible and invisible) region, and the like.

[0193] The reference signal may be abbreviated as RS (reference signal), or may be referred to as a pilot according to an applied standard.

[0194] As used in this disclosure, the phrase “based on” does not mean “based only on”, unless explicitly stated otherwise. In other words, the description “based on” means both “based only on” and “based at least on”.

[0195] Any reference to elements using designations such as “first” and “second” as used in the present disclosure does not generally limit the amount or order of those elements. These designations may be used in the present disclosure as a convenient way to distinguish between two or more elements. Thus, references to first and second elements do not imply that only two elements may be employed or that the first element must in any way precede the second element.

[0196] The “means” in the configuration of each device described above may be replaced with a “unit”, a “circuit”, a “device”, or the like.

[0197] When the present disclosure uses the terms “include”, “including”, and variations thereof, these terms are intended to be inclusive in a manner similar to the term “comprising”. Furthermore, the term “or” used in the present disclosure is intended not to be an exclusive OR.

[0198] A radio frame may be configured with one or a plurality of frames in a time domain. Each of the one or a plurality of frames in the time domain may be referred to as a sub-frame. The sub-frame may further be configured with one or a plurality of slots in the time domain. The sub-frame may be a fixed time length (for example, 1 ms) that is independent of numerology.

[0199] The numerology may be a communication parameter applied to at least one of transmission or reception of a signal or a channel. The numerology may indicate at least one of, for example, a sub-carrier spacing (SCS), a bandwidth, a symbol length, a cyclic prefix length, a transmission time interval (TTI), the number of symbols per TTI, a radio frame configuration, a particular filtering operation performed by a transceiver in the frequency domain, a particular windowing operation performed by the transceiver in the time domain, or the like.

[0200] The slot may be configured with one or a plurality of symbols (an orthogonal frequency division multiplexing (OFDM) symbol, a single carrier frequency division multiple access (SC-FDMA) symbol, and the like) in the time domain. The slot may be a time unit based on the numerology.

[0201] The slot may include a plurality of mini-slots. Each mini-slot may be configured with one or a plurality of symbols in the time domain. Further, the mini-slot may be referred to as a sub-slot. The mini-slot may be configured with a smaller number of symbols than the slot. A PDSCH (or PUSCH) transmitted in a time unit larger than the mini-slot may be referred to as a PDSCH (or PUSCH) mapping type A. A PDSCH (or PUSCH) transmitted using the mini-slot may be referred

to as a PDSCH (or PUSCH) mapping type B.

[0202] Each of the radio frame, the sub-frame, the slot, the mini-slot, and the symbol represents a time unit when a signal is transmitted. Different names respectively corresponding to the radio frame, the sub-frame, the slot, the mini-slot, and the symbol may be used.

[0203] For example, one sub-frame may be referred to as a transmission time interval (TTI), a plurality of consecutive sub-frames may be referred to as a TTI, and one slot or one mini-slot may be referred to as a TTI. That is, at least one of the sub-frame or the TTI may be a sub-frame (1 ms) in existing LTE, a period of time shorter than 1 ms (for example, 1 to 13 symbols), or a period of time longer than 1 ms. It is noted that the unit representing the TTI may be referred to as the slot, the mini-slot, or the like instead of the sub-frame.

[0204] Here, the TTI refers to, for example, a minimum time unit of scheduling in wireless communication. For example, in the LTE system, the base station performs scheduling of allocating radio resources (frequency bandwidth, transmission power, and the like that can be used in each terminal **20**) to each terminal **20** in units of TTIs. It is noted that the definition of the TTI is not limited thereto.

[0205] The TTI may be a transmission time unit such as a channel coded data packet (a transport block), a code block, or a code word, or may be a processing unit such as scheduling or link adaptation. It is noted that, when a TTI is given, a time interval (for example, the number of symbols) in which the transport block, the code block, the code word, and the like are actually mapped may be shorter than the TTI.

[0206] It is noted that, when one slot or one mini-slot is referred to as the TTI, one or more TTIs (that is, one or more slots or one or more mini-slots) may be the minimum time unit of scheduling. Furthermore, the number of slots (the number of mini-slots) constituting the minimum time unit of the scheduling may be controlled.

[0207] A TTI with a time length of 1 ms may be referred to as a general TTI (TTI in LTE Rel. 8-12), a normal TTI, a long TTI, a general sub-frame, a normal sub-frame, a long sub-frame, a slot, or the like. A TTI shorter than the general TTI may be referred to as a shortened TTI, a short TTI, a partial TTI (or fractional TTI), a shortened sub-frame, a short sub-frame, a mini-slot, a sub-slot, a slot, or the like.

[0208] It is noted that the long TTI (for example, the general TTI, the sub-frame, or the like) may be replaced with a TTI having a time length exceeding 1 ms, and the short TTI (for example, the shortened TTI or the like) may be replaced with a TTI having a TTI length less than the TTI length of the long TTI and equal to or greater than 1 ms.

[0209] The resource block (RB) is a resource allocation unit in the time domain and the frequency domain, and may include one or a plurality of consecutive sub-carriers in the frequency domain. The number of sub-carriers included in the RB may be the same regardless of the numerology, for example, may be 12. The number of sub-carriers included in the RB may be determined based on the numerology.

[0210] In addition, the time domain of the RB may include one or a plurality of symbols, and may be a length of one slot, one mini-slot, one sub-frame, or one TTI. Each of one TTI, one sub-frame, and the like may be configured with one or a plurality of resource blocks.

[0211] It is noted that one or a plurality of RBs may be referred to as a physical resource block (physical RB, PRB), a sub-carrier group (SCG), a resource element group (REG), a PRB pair, an RB pair, or the like.

[0212] Furthermore, the resource block may include one or a plurality of resource elements (REs). For example, one RE may be a radio resource area of one sub-carrier and one symbol.

[0213] A bandwidth part (BWP) (which may also be referred to as a partial bandwidth or the like) may represent a subset of contiguous common resource blocks (RBs) for a numerology in a carrier. Here, the common RB may be specified by an index of the RB based on the common reference point of the carrier. The PRB may be defined by a certain BWP and numbered within the BWP.

[0214] The BWP may include a BWP for UL (UL BWP) and a BWP for DL (DL BWP). One or a plurality of BWPs may be configured in one carrier for the UE.

[0215] At least one of the configured BWPs may be active, and the UE may not assume that a predetermined signal/channel is transmitted and received outside the active BWPs. It is noted that “cell”, “carrier”, and the like in the present disclosure may be replaced with “BWP”.

[0216] The above-described structures such as the radio frame, the sub-frame, the slot, the mini-slot, and the symbol are merely examples. For example, the number of sub-frames included in the radio frame, the number of slots per sub-frame or radio frame, the number of mini-slots included in the slot, the number of symbols and RBs included in the slot or mini-slot, the number of sub-carriers included in the RB, the number of symbols in the TTI, a symbol length, a cyclic prefix (CP) length, and the like can be variously changed.

[0217] In the present disclosure, for example, when articles such as a, an, and the in English are added by translation, the present disclosure may include a case in which a noun following these articles is a plural form.

[0218] In the present disclosure, the term “A and B are different” may mean “A and B are different from each other”. It is noted that the term may mean that “A and B are different from C”. Terms such as “separated from” and “coupled to” may be interpreted in the same manner as “different”.

[0219] Each aspect/embodiment described in the present disclosure may be used alone, or may be used in combination, or may be used by being switched with execution. Furthermore, notification of predetermined information (for example, notification of “being X”) is not limited to being performed explicitly, and may be performed implicitly (for example, the predetermined information is not notified).

[0220] Although the present disclosure has been described in detail above, it is apparent to those skilled in the art that the present disclosure is not limited to the embodiments described in the present disclosure. The present disclosure can be implemented as modifications and variations without departing from the spirit and scope of the present disclosure defined by the claims. Therefore, the description of the present disclosure is given for the purpose of illustration and does not have any restrictive meaning to the present disclosure.

REFERENCE SIGNS LIST

[0221] **10** Base station [0222] **110** Transmitting unit [0223] **120** Receiving unit [0224] **130** Configuring unit [0225] **140** Control unit [0226] **20** Terminal [0227] **210** Transmitting unit [0228] **220** Receiving unit [0229] **230** Configuring unit [0230] **240** Control unit [0231] **1001** Processor [0232] **1002** Storage device [0233] **1003** Auxiliary storage device [0234] **1004** Communication device [0235] **1005** Input device [0236] **1006** Output device [0237] **2001** Vehicle [0238] **2002** Drive unit [0239] **2003** Steering unit [0240] **2004** Accelerator pedal [0241] **2005** Brake pedal [0242] **2006** Shift lever [0243] **2007** Front wheel [0244] **2008** Rear wheel [0245] **2009** Axle [0246] **2010** Electronic control unit [0247] **2012** Information service unit [0248] **2013** Communication module [0249] **2021** Current sensor [0250] **2022** Rotation speed sensor [0251] **2023** Air pressure sensor [0252] **2024** Vehicle speed sensor [0253] **2025** Acceleration sensor [0254] **2026** Brake pedal sensor [0255] **2027** Shift lever sensor [0256] **2028** Object detection sensor [0257] **2029** Accelerator pedal sensor [0258] **2030** Driving assistance system unit [0259] **2031** Microprocessor [0260] **2032** Memory (ROM, RAM) [0261] **2033** Communication port (IO port)

Claims

1. A terminal comprising: a receiving unit configured to receive, from a base station, a downlink shared channel in a virtual CC (component carrier) that is discontinuous in a frequency domain and that includes a plurality of actual CCs; a control unit configured to determine feedback information related to retransmission control corresponding to the downlink shared channel; and a transmitting unit configured to transmit, to the base station, an uplink control channel including the feedback

information in the virtual CC, wherein the control unit determines which actual CC included in the virtual CC transmits the uplink control channel, and determines an interval between the downlink shared channel and the uplink control channel.

2. The terminal according to claim 1, wherein the control unit determines the interval between the downlink shared channel and the uplink control channel in an actual CC receiving the downlink shared channel.

3. The terminal according to claim 1, wherein the control unit determines the interval between the downlink shared channel and the uplink control channel in an actual CC transmitting the uplink control channel.

4. The terminal according to claim 1, wherein the control unit determines an interval between a head of the downlink shared channel and the uplink control channel.

5. The terminal according to claim 1, wherein the control unit determines an interval between a tail end of the downlink shared channel and the uplink control channel.

6. A communication method to be performed by a terminal, comprising: a reception procedure of receiving, from a base station, a downlink shared channel in a virtual CC (component carrier) that is discontinuous in a frequency domain and that includes a plurality of actual CCs; a control procedure of determining feedback information related to retransmission control corresponding to the downlink shared channel; a transmission procedure of transmitting, to the base station, an uplink control channel including the feedback information in the virtual CC; and a procedure of determining which actual CC included in the virtual CC transmits the uplink control channel and determining an interval between the downlink shared channel and the uplink control channel.
